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THE COLLEGE OF
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DEPARTMENT OF
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SOFTWARE-BASED STOLEN VEHICLE IDENTIFICATION AND ALERT SYSTEM (SSVIAS) FOR BAMENDA, CAMEROON

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BY:

Name

MBOHBENEH SIMPLICE QUOTEKA

Registration Number:

UBa23PH026

SUPERVISOR:

Mr MBIETHIEU CEZAR

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ABSTRACT

Vehicle theft remains a growing concern in many urban and semi-urban areas, particularly in developing regions such as Bamenda, Cameroon. With the increasing number of vehicles and limited surveillance infrastructure, tracking and recovering stolen vehicles has become a significant challenge for both vehicle owners and law enforcement authorities. This project presents the design and implementation of a software-based stolen vehicle identification and alert system aimed at enhancing vehicle tracking and recovery in Bamenda.

The system is developed using PHP (Laravel), HTML, CSS, and JavaScript, and provides features such as user authentication, vehicle registration, stolen vehicle reporting, vehicle verification using plate numbers, image-based number plate recognition using Optical Character Recognition (OCR) technology, and real-time notification alerts. The application allows vehicle owners to register and manage their vehicle information digitally, report stolen vehicles instantly, and receive alerts. Members of the public and law enforcement officers can verify whether a vehicle is stolen using the web platform.

The proposed solution improves the speed and efficiency of stolen vehicle identification, enhances collaboration between vehicle owners, the public, and law enforcement, and provides a centralized platform for vehicle data management. The system is designed to operate effectively under intermediate internet conditions common in Bamenda, without relying on expensive hardware such as sensors or surveillance cameras.

DEDICATION

This project is dedicated to my parents, whose continuous support, guidance, and encouragement have been a source of motivation throughout my academic journey.

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LIST OF ABBREVIATIONS

Abbreviation	Meaning
API	Application Programming Interface
CSS	Cascading Style Sheets
GPS	Global Positioning System
HTML	HyperText Markup Language
HTTP	HyperText Transfer Protocol
IoT	Internet of Things
MVC	Model-View-Controller
NGO	Non-Governmental Organization
OCR	Optical Character Recognition
PHP	Hypertext Preprocessor
SQL	Structured Query Language
UI	User Interface
URL	Uniform Resource Locator
UX	User Experience

CHAPTER ONE

INTRODUCTION

1.1 Context (Background of the Study)

Vehicle theft is a growing concern in many urban and semi-urban areas, particularly in developing regions such as Bamenda, Cameroon. With the increasing number of vehicles and limited surveillance infrastructure, tracking and recovering stolen vehicles has become a significant challenge for both vehicle owners and law enforcement authorities. In many parts of the North West Region, vehicles are essential for daily transportation, commerce, and livelihoods, making vehicle theft not only a financial loss but also a disruption to community life.

In Bamenda, the process of identifying and recovering stolen vehicles is largely manual and inefficient. Victims often rely on police reports, word-of-mouth communication, and informal checkpoints, which are slow and unreliable. There is no centralized digital platform that allows for real-time reporting, verification, and tracking of stolen vehicles. This lack of coordination often results in delayed responses and significantly reduces the chances of recovering stolen vehicles.

With the rapid adoption of smartphones and web technologies, there is an opportunity to leverage software-based systems to address this problem. A web-based stolen vehicle identification system can provide a centralized platform where vehicle data is stored, stolen vehicles are reported, and users can verify vehicle status in real time. Such a system can bridge the gap between vehicle owners, the public, and law enforcement authorities by enabling timely communication and data sharing.

This project focuses on the design and implementation of a software-based stolen vehicle identification and alert system, tailored to the local conditions of Bamenda, without relying on hardware such as sensors or surveillance cameras. The system aims to provide an affordable, accessible, and scalable solution that leverages existing web technologies and community participation.

1.2 Problem Statement

Despite the increasing rate of vehicle theft in Bamenda, there is no efficient and centralized system for reporting, tracking, and identifying stolen vehicles. The current methods are

fragmented, slow, and largely manual, making it difficult to recover stolen vehicles in a timely manner.

Vehicle owners often lack immediate means to alert the public or authorities when theft occurs. Similarly, members of the public and law enforcement officers have no easy way to verify whether a vehicle is stolen. This results in delayed responses and reduced chances of recovery. The absence of a centralized database means that stolen vehicle information is not readily accessible to those who need it most.

Additionally, existing advanced solutions such as surveillance systems and automated detection technologies are expensive and require specialized hardware, making them unsuitable for local deployment in many communities. These systems often demand significant infrastructure investment, which is not feasible for widespread adoption in developing regions.

Therefore, the key problem addressed in this study is the lack of an affordable, accessible, and software-based system for identifying and reporting stolen vehicles in Bamenda.

1.3 Research Questions

1.3.1 General Research Question

How can a software-based application be used to improve the identification and recovery of stolen vehicles in Bamenda?

1.3.2 Specific Research Questions

- How can vehicle owners register vehicle information and report stolen vehicles through a web-based platform?
- How can users and law enforcement verify the status of a vehicle using number plate recognition technology?
- How can real-time alerts and community participation enhance the detection and recovery of stolen vehicles?

1.4 Objectives of the Study

1.4.1 General Objective

To design and implement a software-based stolen vehicle identification and alert system that enhances vehicle tracking and recovery in Bamenda, Cameroon.

1.4.2 Specific Objectives

- To develop a web-based platform that enables vehicle registration and stolen vehicle reporting by vehicle owners.
- To implement a vehicle verification system using number plate recognition technology for status verification.
- To design a real-time alert mechanism that facilitates community participation in stolen vehicle detection.

1.5 Justification and Motivation

This project is motivated by the increasing cases of vehicle theft and the lack of effective digital solutions to address the problem in Bamenda. Existing approaches are either manual or require expensive hardware systems that are not feasible for widespread use in local communities. Many vehicle owners suffer significant financial losses due to theft, and the lack of an efficient recovery system exacerbates the problem.

By leveraging web technologies, this project provides a low-cost and scalable solution that enables real-time communication, data sharing, and community participation in detecting stolen vehicles. The system empowers ordinary citizens to contribute to vehicle recovery efforts while providing law enforcement with the tools they need to respond effectively.

1.6 Significance of the Study

The proposed system will:

- Improve the speed and efficiency of stolen vehicle identification
- Enhance collaboration between vehicle owners, the public, and law enforcement
- Provide a centralized platform for vehicle data management
- Increase the chances of recovering stolen vehicles
- Serve as a foundation for future smart security systems
- Reduce the reliance on expensive hardware-based solutions

1.7 Scope

This project focuses on the development of a software-based stolen vehicle detection system that operates through a web application. It includes:

- Vehicle registration and management
- Stolen vehicle reporting

- Vehicle verification using plate numbers
- Image-based plate recognition (OCR)
- Notification and alert system

The project does not include hardware-based detection systems such as cameras, sensors, or GPS tracking devices. The scope is limited to software development and web-based functionality suitable for deployment in Bamenda.

1.8 Limitations

One of the major limitations of this study is its dependence on internet connectivity, which may be inconsistent in some parts of Bamenda and surrounding communities, thereby affecting real-time access to the web application. In areas with poor network coverage, users may experience delays in reporting or verifying stolen vehicles.

Additionally, the accuracy of the system depends on user-provided data, which may sometimes be incomplete or inaccurate. The OCR component may also be affected by poor image quality, lighting conditions, or unclear number plates. Users who upload low-quality images may receive inaccurate recognition results. Since the system does not include automated surveillance, it relies heavily on user participation and reporting, which may vary based on community engagement levels.

1.9 Delimitations

This study is deliberately limited in scope to Bamenda and its surrounding communities in the North West Region of Cameroon to ensure feasibility at the undergraduate level. The project focuses strictly on a software-based solution and excludes hardware-based detection systems such as GPS trackers, cameras, or physical sensors.

The system addresses vehicle identification and reporting rather than physical recovery operations or law enforcement investigation procedures. These delimitations allow the study to concentrate on developing a practical and scalable digital solution while providing a foundation for future expansion that could include hardware integration or broader geographical coverage.

SUMMARY

This project designed and implemented a web-based stolen vehicle identification and alert system to address vehicle theft in Bamenda, Cameroon, where manual reporting methods have hindered recovery efforts. The system enables vehicle registration and theft reporting, integrates OCR technology for number plate verification, and provides real-time alerts to encourage community participation. It offers an affordable, software-based alternative to expensive hardware solutions, bridging the gap between vehicle owners, the public, and law enforcement. Despite limitations such as internet dependency and reliance on user-provided data quality, the system demonstrates that accessible web technology can effectively address local security challenges. This project serves as a practical foundation for future enhancements, with the potential to improve vehicle recovery rates and contribute to safer communities in Bamenda.

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